Advanced information, computation, communication I
EPFL - Fall semester 2026-2027

Week 7

October 30, 2026

Exercise 1. The Selection Sort algorithm sorts the input sequence $a_1, ..., a_n$ of real numbers in ascending order

Algorithm 1 Selection Sort

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1: for i = 1 to n − 1 do
2:     min ← i + 1
3:     for j = i + 1 to n do
4:         if a_min > a_j then
5:             min ← j
6:         end if
7:     end for
8:     if a_i > a_min then
9:         swap a_i and a_min
10:    end if
11: end for
```

Which sequence will result in the most operations when using Selection Sort (swaps and comparisons)?

- ◯ 9, 3, 8, 6, 5, 1, 7
- ◯ 4, 7, 2, 9, 5, 6, 3
- ◯ 0, 1, 2, 3, 4, 5, 6
- ◯ 7, 6, 5, 4, 3, 2, 1

Exercise 2.

$$L_{x_1} = (y_3, y_1, y_2) \qquad L_{y_1} = (x_2, x_1, x_3)$$
$$L_{x_2} = (y_2, y_3, y_1) \qquad L_{y_2} = (x_1, x_3, x_2)$$
$$L_{x_3} = (y_1, y_2, y_3) \qquad L_{y_3} = (x_3, x_2, x_1)$$

Let L_x for $x \in X = \{x_1, x_2, x_3\}$ be the preference list of x as given above
and let L_y for $y \in Y = \{y_1, y_2, y_3\}$ be the preference list of y as given above.
We say that a matching is X-optimal (resp. Y-optimal) if all elements of X (resp. Y) are matched with their highest preference.
The matching $\{(x_1, y_1), (x_2, y_3), (x_3, y_2)\}$ is

- ◯ unstable.
- ◯ stable and Y-optimal.
- ◯ stable and X-optimal.
- ◯ stable but not a stable matching that is X- or Y-optimal.

Exercise 3. Use the cashier's algorithm to make change using quarters (25 cents), dimes (10 cents), nickels (5 cents), and pennies (1 cent) for:

1. 87 cents.
2. 49 cents.
3. 99 cents.
4. 33 cents.

Exercise 4. Let $\{A, B, C, D\}$ be a set of men, and $\{a, b, c, d\}$ a set of women. We want to match up men and women using the Gale-Shapley algorithm in two different ways. The preferences of men and women are given in the following lists, going from most preferable on the left to least preferable on the right.

Men	1st	2nd	3rd	4th
A	c	d	b	a
B	d	c	a	b
C	a	c	b	d
D	b	d	a	c

Women	1st	2nd	3rd	4th
a	D	A	B	C
b	C	B	A	D
c	C	B	A	D
d	D	A	B	C

1. If the men propose, and women accept/reject, what is the matching after the algorithm terminates?
2. If the women propose, and men accept/reject, what is the matching after the algorithm terminates?
3. Who is the best possible (stable) valid partner for "a"?

Exercise 5. Which function below grows fastest when n goes to infinity?

- ◯ $(\log_3(33))^{n-3}$
- ◯ 3^n
- ◯ $n^{3\log_3(n)}$
- ◯ $n^3 \log_3(n)$

Exercise 6. Suppose that $f_1(x)$ is $O(g_1(x))$ and that $f_2(x)$ is $O(g_2(x))$.
Show that $(f_1 + f_2)(x)$ is $O(\max(|g_1(x)|, |g_2(x)|))$:

Exercise 7. [**Exam 2023**] Order the following functions so that each function is big-O of the next function. For example if f is $O(g)$ and g is $O(h)$, the order should be f, g, h. Justify your choices.

$$
\begin{aligned}
f_1(n) &= 2^{100000} \\
f_2(n) &= \left(\log_2(n^2)\right)^2 \\
f_3(n) &= 0.001n! \\
f_4(n) &= n^{0.01}
\end{aligned}
$$